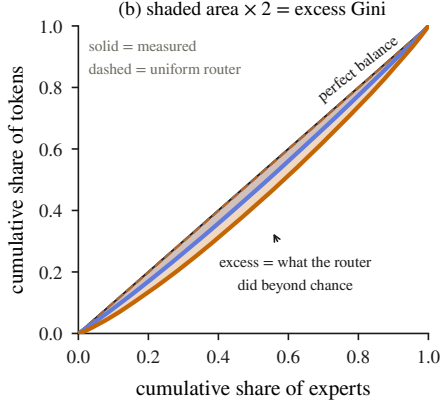
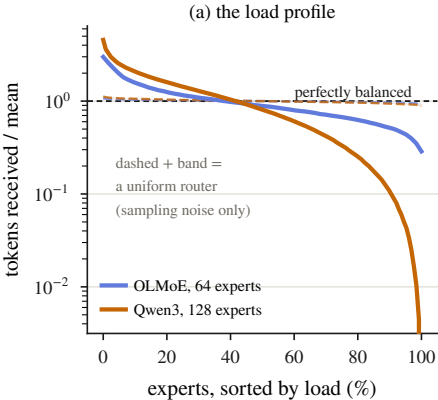




Gini alone is unreadable: a uniform router scores 0.51 at 1 token per expert and 0.02 at 1000, purely from multinomial noise. So each measurement is compared with that null -- the same token count thrown uniformly over the same experts -- and the EXCESS is the part the router is responsible for. Both panels are prefill B=64 on ShareGPT, the one setting where the two models are directly comparable. OLMoE: gini 0.24, null 0.02, excess +0.22. Qwen3: gini 0.46, null 0.03, excess +0.44 -- twice the imbalance, on identical prompts.